

HOMEWORK 9
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

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SECTION 5.4 - EIGENVECTORS AND LINEAR TRANSFORMATIONS

1. Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$ be the transformation that maps a polynomial, $p(t)$, to its derivative, $p'(t)$. Show that T is a linear transformation. Then, find the matrix for T relative to the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \{1, t\}$ for $\mathbb{P}_1$.

As $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$ takes a polynomial $p(t) \in \mathbb{P}_2$ to its derivative $p'(t) \in \mathbb{P}_1$, let's see what T does to a generic polynomial $p(t) = a + bt + ct^2 \in \mathbb{P}_2$ where $a, b, c \in \mathbb{R}$ are arbitrary constants. That is, let's observe that

$$T(p(t)) = \frac{d}{dt}(a + bt + ct^2) = b + 2ct.$$

So, using the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \{1, t\}$ for $\mathbb{P}_1$, we have that the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$ is the matrix M such that if

$$[p(t)]_{\mathcal{B}} = [a + bt + ct^2]_{\mathcal{B}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

then

$$M[p(t)]_{\mathcal{B}} = [T(p(t))]_{\mathcal{D}} = [b + 2ct]_{\mathcal{D}} = \begin{bmatrix} b \\ 2c \end{bmatrix}.$$

Using the formula for M provided in Section 5.4 of the textbook, we have

$$\begin{aligned} M &= \begin{bmatrix} [T(b_1)]_{\mathcal{D}} & [T(b_2)]_{\mathcal{D}} & [T(b_3)]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} [T(1)]_{\mathcal{D}} & [T(t)]_{\mathcal{D}} & [T(t^2)]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} [0]_{\mathcal{D}} & [1]_{\mathcal{D}} & [2t]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \end{aligned}$$

Let's check that this actually works by confirming that $M[p(t)]_{\mathcal{B}} = [T(p(t))]_{\mathcal{D}}$ for an arbitrary polynomial $p(t) = a + bt + ct^2 \in \mathbb{P}_2$:

$$\begin{aligned} M[p(t)]_{\mathcal{B}} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \\ &= \begin{bmatrix} b \\ 2c \end{bmatrix} \\ &= [T(p(t))]_{\mathcal{D}}. \end{aligned}$$

So, this agrees with our previous calculation, and we may conclude that the matrix M given above is indeed the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$.

2. The *trace* of an $n \times n$ matrix A is the sum of the diagonal entries in A and is denoted $tr(A)$. If B is another $n \times n$ matrix, can be shown that $tr(AB) = tr(BA)$. Use this fact to show that if A and B are similar, then $tr(A) = tr(B)$.

Let A and B be $n \times n$ matrices and suppose that A and B are similar. Then, by the definition of similarity, we have $A = PBP^{-1}$ for some invertible $n \times n$ matrix P . Using the fact that $tr(AB) = tr(BA)$ as given in the statement of the problem, this tells us that

$$\begin{aligned} tr(A) &= tr(PBP^{-1}) \\ &= tr((PB)P^{-1}) \\ &= tr(P^{-1}(PB)) \\ &= tr((P^{-1}P)B) \\ &= tr(IB) \\ &= tr(B). \end{aligned}$$

So, we may indeed conclude that if A and B are similar $n \times n$ matrices, then $tr(A) = tr(B)$.

3. Show that the trace of a diagonalizable matrix A is equal to the sum of the eigenvalues of A . (While we won't prove it here, this is actually true for *any* matrix, regardless of whether or not it is diagonalizable.)

Let A be a diagonalizable matrix. Then, by the definition of diagonalizability, we have $A = PDP^{-1}$ for some invertible matrix P and diagonal matrix D .

Using our result above, we know that $\text{tr}(A) = \text{tr}(D)$, since A and D are similar matrices. By the Diagonalization Theorem (Theorem 5 of section 5.3 in your textbook), the diagonal entries of D are the eigenvalues of A . So, as the trace of a matrix M is, by definition, the sum of the diagonal entries, the trace of D is the sum of the eigenvalues of A .

Hence, as $\text{tr}(A) = \text{tr}(D)$, we may conclude that the trace of A is the sum of the eigenvalues of A .

SECTION 5.5 - COMPLEX EIGENVALUES

5. Let the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$ act on $\mathbb{C}^2$. Find the eigenvalues of A and find a basis for each eigenspace of A in $\mathbb{C}^2$.

To find the eigenvalues of A , we must first find the characteristic polynomial of A :

$$\det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & -2 \\ 1 & 3-\lambda \end{bmatrix} \right) = \lambda^2 - 4\lambda + 5.$$

To find the eigenvalues of A , we simply solve $\det(A - \lambda I) = 0$ for λ . Using the above, we have

$$\lambda^2 - 4\lambda + 5 = (\lambda - (2 - i))(\lambda - (2 + i)).$$

So, the eigenvalues of A are $\lambda_1 = 2 - i$ and $\lambda_2 = 2 + i$ (notice, the eigenvalue λ_1 of A is the complex conjugate of the eigenvalue λ_2 and vice versa). To find a basis for each eigenspace, we need only determine a basis (in $\mathbb{C}^2$) for $\text{Nul}(A - \lambda I)$ where $\lambda = \lambda_1$ and $\lambda = \lambda_2$. So, we begin by constructing

$$A - \lambda_1 I = \begin{bmatrix} 1 - (2 - i) & -2 \\ 1 & 3 - (2 - i) \end{bmatrix} = \begin{bmatrix} -1 + i & -2 \\ 1 & 1 + i \end{bmatrix}.$$

Then, we solve $(A - \lambda_1 I)x = 0$. This can be done in the same fashion as usual, but now our algebra will range over $\mathbb{C}$ instead of $\mathbb{R}$.

$$\begin{aligned} \begin{bmatrix} -1 + i & -2 \\ 1 & 1 + i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} (-1 + i)x_1 - 2x_2 \\ x_1 + (1 + i)x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

From the above, we can see that this results in the system

$$\begin{aligned} (-1 + i)x_1 - 2x_2 &= 0 \\ x_1 + (1 + i)x_2 &= 0. \end{aligned}$$

From the second equality, $x_2 = \frac{-1}{1+i}x_1$ must hold. Substituting into the first equality, this produces the equation $(-1 + i)x_1 + \frac{2}{1+i}x_1 = 0$. Solving, we have

$$\begin{aligned} (-1 + i)x_1 + \frac{2}{1+i}x_1 &= 0 \\ \left((-1 + i) + \frac{2}{1+i} \right)x_1 &= 0 \\ ((-1 + i)(1 + i) + 2)x_1 &= 0 \\ (0)x_1 &= 0. \end{aligned}$$

So, since this is true for any value of x_1 , we have that the solution to $(A - \lambda_1 I)x = 0$ is given by any vector of the form $x = x_1 \begin{bmatrix} 1 \\ \frac{-1}{1+i} \end{bmatrix} = \frac{x_1}{1+i} \begin{bmatrix} 1+i \\ -1 \end{bmatrix}$. Hence, a basis for the eigenspace of A associated to the eigenvalue $\lambda_1 = 2 - i$ is $\left\{ \begin{bmatrix} 1+i \\ -1 \end{bmatrix} \right\}$. Since eigenvalues come in conjugate pairs, and as the corresponding eigenvectors associated to these conjugate eigenvalues are themselves conjugate, a basis for the eigenspace of A associated to the eigenvalue $\lambda_2 = 2 + i$ is $\left\{ \begin{bmatrix} 1-i \\ -1 \end{bmatrix} \right\}$.

6. Let A be an $n \times n$ real matrix, and let $x \in \mathbb{C}^n$. Show that

$$\operatorname{Re}(Ax) = A(\operatorname{Re}(x)) \quad \text{and} \quad \operatorname{Im}(Ax) = A(\operatorname{Im}(x)).$$

To begin, notice that every complex vector $x \in \mathbb{C}^n$ may be written as

$$x = \operatorname{Re}(x) + i \operatorname{Im}(x).$$

That is, a complex vector (just like a complex number) consists of a real part plus an imaginary part multiplied by the complex number i . So, by the linearity of matrix multiplication, we have

$$\begin{aligned} Ax &= A(\operatorname{Re}(x) + i \operatorname{Im}(x)) \\ &= A(\operatorname{Re}(x)) + i A(\operatorname{Im}(x)). \end{aligned}$$

From here, the desired result is easy to obtain. Computing $\operatorname{Re}(Ax)$ using the above expansion, we have:

$$\begin{aligned} \operatorname{Re}(Ax) &= \operatorname{Re}(A(\operatorname{Re}(x)) + i A(\operatorname{Im}(x))) \\ &= A(\operatorname{Re}(x)). \end{aligned}$$

Computing $\operatorname{Im}(Ax)$ using the above expansion, we have:

$$\begin{aligned} \operatorname{Im}(Ax) &= \operatorname{Im}(A(\operatorname{Re}(x)) + i A(\operatorname{Im}(x))) \\ &= A(\operatorname{Im}(x)). \end{aligned}$$

So, as shown by the above calculations, we may indeed conclude that for any real $n \times n$ matrix A and any vector $x \in \mathbb{C}^n$, we have $\operatorname{Re}(Ax) = A(\operatorname{Re}(x))$ and $\operatorname{Im}(Ax) = A(\operatorname{Im}(x))$.

SECTION 5.7 - APPLICATIONS TO DIFFERENTIAL EQUATIONS

5. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$. Determine whether the origin is a stable or unstable equilibrium point of the system $x'(t) = Ax(t)$.

First, we calculate the eigenvalues of A :

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ \det\left(\begin{bmatrix} 2-\lambda & 3 \\ -1 & -2-\lambda \end{bmatrix}\right) &= 0 \\ (2-\lambda)(-2-\lambda) - (3)(-1) &= 0 \\ \lambda^2 - 1 &= 0 \\ (\lambda - 1)(\lambda + 1) &= 0\end{aligned}$$

So, the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = -1$. Routine calculation shows that the eigenvectors corresponding to these eigenvalues are $v_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. The general form of a solution to the linear system $x'(t) = Ax(t)$ is then given by

$$x(t) = c_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t}$$

where $c_1, c_2 \in \mathbb{R}$ are arbitrary constants.

Using this, it's easy to see that the origin is an *unstable equilibrium point* of this system, since $e^t \rightarrow \infty$ as $t \rightarrow \infty$. That is, if $c_1 \neq 0$, then $x(t)$ will not approach the origin as $t \rightarrow \infty$.

In fact, the origin is a *saddle point* (also known as a *conditionally stable equilibrium point*) of the system, as trajectories $x(t)$ for the system will converge to the origin if $c_1 = 0$ and will exponentially diverge from the origin if $c_1 \neq 0$. Further details regarding this phenomenon may be found in section 5.7 of your textbook.

6. Let A be as in problem 5. Solve the initial value problem

$$\begin{cases} x'(t) = Ax(t) \\ x(0) = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{cases}$$

Using your answer to problem 5, is it true that the solution you found satisfies

$$\lim_{t \rightarrow \infty} x(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} ?$$

Using our solution to problem 5, we know that

$$x(t) = c_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t}$$

is the general form of the solution to the dynamical system $x'(t) = Ax(t)$. Since we are looking for the *particular solution* $x(t)$ which satisfies

$$\begin{cases} x'(t) = Ax(t) \\ x(0) = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{cases}$$

we need to determine the values of c_1 and c_2 so that $x(0) = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. This amounts to solving

$$\begin{aligned} x(0) &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ c_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^0 + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-0} &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ \begin{bmatrix} -3c_1 - c_2 \\ c_1 + c_2 \end{bmatrix} &= \begin{bmatrix} 3 \\ 2 \end{bmatrix}. \end{aligned}$$

This induces the corresponding system of equations

$$\begin{aligned} -3c_1 - c_2 &= 3 \\ c_1 + c_2 &= 2. \end{aligned}$$

From the second equation, we can see that $c_2 = 2 - c_1$. Substituting this into the first equation, we produce

$$\begin{aligned} -3c_1 - (2 - c_1) &= 3 \\ -2c_1 &= 5 \\ c_1 &= -\frac{5}{2}. \end{aligned}$$

Correspondingly, we may then see that $c_2 = \frac{9}{2}$. So, the solution to the initial value problem given above is

$$x(t) = -\frac{5}{2} \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^t + \frac{9}{2} \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t} = \begin{bmatrix} (15/2)e^t - (9/2)e^{-t} \\ -(5/2)e^t + (9/2)e^{-t} \end{bmatrix}.$$

Since $\lim_{t \rightarrow \infty} e^t = \infty$, it then follows that

$$\lim_{t \rightarrow \infty} x(t) \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$